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# Electric Propulsion Mechanical Design Intern

San Jose, California, United States

Apply

Archer is an aerospace company based in San Jose, California building an all-electric vertical takeoff and landing aircraft with a mission to advance the benefits of sustainable air mobility. We are designing, manufacturing, and operating an all-electric aircraft that can carry four passengers while producing minimal noise.

Our sights are set high and our problems are hard, and we believe that diversity in the workplace is what makes us smarter, drives better insights, and will ultimately lift us all to success. We are dedicated to cultivating an equitable and inclusive environment that embraces our differences, and supports and celebrates all of our team members.

This internship is a part of a 10-week full summer internship program opportunity role will be based at Archer's headquarters in

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This internship is a part of a 10-week full-time summer internship program opportunity. This role will be based at Archer's headquarters in San Jose, California and relocation or housing will not be provided.

Archer is seeking a motivated and detail-oriented intern to join Archer. This internship provides a hands-on opportunity to gain practical experience within the Electric Propulsion team. In this role, you will support the design, development, and validation of Archer's high-performance electric propulsion units. You will play a vital role in bridging the gap between digital design and physical hardware, helping the team optimize the electric motor, gearbox, and tilt mechanism assemblies for flight-critical safety and weight targets.

**What You'll Do:**

- **Component Design:** Design and model complex mechanical components for electric motors and gearboxes (such as housings, structural components, cooling jackets, stator, and rotor assemblies) using 3D CAD software, focusing on mass optimization and manufacturability.
- **Engineering Documentation:** Ge
detailed 2D component drawings a

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- **Engineering Documentation:** Generate detailed 2D component drawings and assembly instructions utilizing Geometric Dimensioning and Tolerancing (GD&T) standards to ensure parts are manufactured and assembled correctly.
- **Test Equipment Development:** Design, model, and source parts for custom test stands, to validate propulsion sub-systems on the ground.
- **Test Execution:** Assist in the hands-on setup and execution of development tests, including installing sensors, running test scripts, and monitoring safety limits during operation.
- **Data Analysis & Reporting:** Post-process raw testing data to evaluate performance against predictions and author technical reports to document results, failures, and recommendations for design improvements.
- **Prototyping & Assembly:** Support the physical build, integration, and teardown of prototype propulsion units, working closely with technicians to identify assembly issues and improve serviceability.
- **Cross-Functional Collaboration:** Collaborate with thermal, magnetic manufacturing engineering teams t ensure designs meet multi-disciplinary

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- **Cross-Functional Collaboration:** Collaborate with thermal, magnetic, and manufacturing engineering teams to ensure designs meet multi-disciplinary requirements and aggressive project timelines.

**What You Need:**

- **Education:** Currently pursuing a Bachelor's degree in Mechanical Engineering, Aerospace Engineering, or Mechatronics (entering 2nd or 3rd year).
- **CAD Proficiency:** Strong experience with parametric 3D CAD modeling (Siemens NX, SolidWorks, or CATIA); experience with Siemens NX is a distinct plus.
- **Drawing Standards:** Fundamental understanding of technical drawing creation and familiarity with GD&T (ASME Y14.5) concepts.
- **Data Processing:** Basic proficiency in Python, MATLAB, or advanced Excel for parsing test data and generating plots/graphs.
- **Hands-on Experience:** Experience with physical prototyping, such as 3D printing, manual machining (mill/lathe), or university competition teams (FSAE, Solar Car, Rocketry).
- **Engineering Fundamentals:** Solid understanding of core mechanical

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- **Engineering Fundamentals:** Solid understanding of core mechanical engineering concepts including statics, dynamics, mechanics of materials, and basic thermodynamics.
- **Communication:** Strong verbal and written communication skills, with the ability to document technical work clearly for a broad audience.
- **Curiosity & Drive:** A demonstrated passion for aviation, electric vehicles, or clean technology, and a willingness to learn in a fast-paced, startup-like environment.

**Summer 2026 Internship Program Details:**

- This internship is a part of a 10-week full-time summer internship program opportunity. Program dates are:
  - May 26, 2026 - July 31, 2026
  - June 15, 2026 - August 21, 2026
    - *We will not be able to accommodate interns outside of these two program dates.
- This role will be based at Archer in San Jose, CA
- Relocation and housing will not be provided.

Please note that this job description is intended to provide a general overview of the

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Please note that this job description is intended to provide a general overview of the position and does not include an exhaustive list of responsibilities and qualifications.

At Archer we aim to attract, retain, and motivate talent that possess the skills and leadership necessary to grow our business. We drive a pay-for-performance culture and reward performance that supports the Company's business strategy. For this position we are targeting a base pay between $25.00 - $35.00 per hour. Actual compensation offered will be determined by factors such as job-related knowledge, skills, and experience.

Archer is committed to working with and providing reasonable accommodations to job applicants with physical or mental disabilities, and those with sincerely held religious beliefs. Applicants who may require reasonable accommodation for any part of the application or hiring process should provide their name and contact information to Archer's People Team at people@archer.com. Reasonable accommodations will be determined on a case-by-case basis.

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